

PlayMates

A Multi-Regional Gaming Community Hub

Web Technology Mini-Project Document

INSTITUTION	MGM University
COURSE / SCOPE	Web Technology Mini-Project
TECHNOLOGY STACK	HTML5, CSS3, Bootstrap, Tailwind CSS
DOCUMENT VERSION	1.0 (Final Academic Revision)

Project Contributors

STUDENT NAME	PRN NUMBER	DIVISION
Parth Chormale	202501103144	SY 3
Viraj Bhosale	202501103141	SY 2

Note: PlayMates is designed as a streamlined, front-end Web Technology project focused on responsive UI/UX and modular design without heavy backend requirements.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the operational and functional requirements of **PlayMates**, a simple web-based gaming community platform. It details the system features, user interaction workflows, visual hierarchy, and structural guidelines of the platform.

1.2 Project Overview

PlayMates consolidates essential gaming activities into a unified, clean web application. Users can discover trending games, read and publish community reviews, receive rule-based game recommendations, find co-players by region/language, join gaming communities, create game rooms, participate in text chats, publish posts, and join esports/gaming events.

1.3 Project Scope

The project is intentionally tailored as a lightweight, realistic Web Technology mini-project. It highlights modern front-end web development practices using HTML5, CSS3, Bootstrap, and Tailwind CSS to ensure responsiveness and interactive presentation.

1.4 Objectives

- Provide a single, intuitive hub to discover games and regional communities.
- Enable registered users to rate, review, and comment on games.
- Facilitate matchmaking by connecting players based on game title, region, and language preferences.
- Provide fundamental community tools including game rooms, live chat interfaces, posts, and event listings.
- Maintain a clean code structure that is straightforward to build, test, and present in academic viva examinations.

2. Existing Problem

Currently, gaming information and community engagement activities are fragmented across multiple disconnected platforms. Gamers often struggle with the following pain points:

- **Fragmented Game Discovery:** Locating suitable and trending games requires searching through multiple independent databases.
- **Scattered Reviews:** High-quality, reliable game reviews and community feedback are spread thinly across various forums and stores.
- **Matchmaking Friction:** Finding local players tailored by specific game titles, geographic region, and spoken language is difficult on general social networks.
- **Dispersed Communities:** Gaming events, clan groups, and discussion forums exist in silos, leading to isolated user experiences.

3. Proposed System

PlayMates solves these challenges by centralizing gaming utilities into one coherent web interface. Users can seamlessly browse games, analyze overall ratings, access simple rule-based recommendations, find co-players, join localized communities, participate in discussion rooms, and track events.

Additionally, an integrated Administrative view allows superusers to manage web content, moderate posts, and oversee user activity efficiently.

4. Main Features

Feature Area	Description & Functionality
1. Landing Page	Hero section introducing PlayMates, popular games, featured communities, and direct access to Login/Register modals.
2. User Authentication	Secure user registration, authentication login forms, and quick session logout controls.
3. User Profile	Personalized user dashboards showing username, geographic region, preferred game titles, and bio summary.
4. Game Discovery	Comprehensive catalog allowing users to search, filter, and sort games by genre, platform, or region.
5. Reviews & Ratings	Interactive 1–5 star rating mechanism, text review submission, and automated average rating computation.
6. Recommendations	Rule-based recommendation logic (e.g., matching preferred genres like FPS to top-rated FPS games) without requiring heavy AI/ML modules.
7. Player Matchmaking	Filterable search directory to connect with co-players based on specific game, region, and primary language.
8. Communities	Hub to create, join, leave, and view member directories for gaming clubs and clans.
9. Game Rooms & Chat	Lobby creation, room joining capabilities, and clean text-chat interaction interfaces.
10. Community Posts	Social feed enabling post creation, like counts, and interactive comment threads.
11. Gaming Events	Event management interface to host, browse, and register for competitive gaming tournaments and casual meetups.
12. Admin Panel	Central dashboard for administrators to oversee platform content, moderate user posts, and manage game listings.

5. User Roles

User Role	Permissions & Access Privileges
Guest	Access public landing page, browse game directory, view public reviews and community listings.
Registered User	

User Role	Permissions & Access Privileges
	Manage user profile, post reviews/ratings, receive recommendations, use matchmaking search, join communities/rooms, post in forums, and register for events.
Admin	Full administrative access to manage users, modify game database entries, moderate community posts, and delete inappropriate content.

6. Functional Requirements

Req ID	Requirement Description
FR-01	User can register a new account with validated credentials.
FR-02	User can securely log in and log out of the platform.
FR-03	User can manage and update personal profile details and bio.
FR-04	User can search and browse the game database with active filters.
FR-05	User can submit star ratings and detailed text reviews for games.
FR-06	System provides rule-based game suggestions based on user interests.
FR-07	User can search for other players by game, region, and language filters.
FR-08	User can create, join, and leave community groups.
FR-09	User can create or enter specialized game rooms.
FR-10	User can send text messages within game rooms.
FR-11	User can create public posts, like content, and post comments.
FR-12	User can browse and register participation for gaming events.
FR-13	Admin can moderate users, posts, reviews, and community groups.

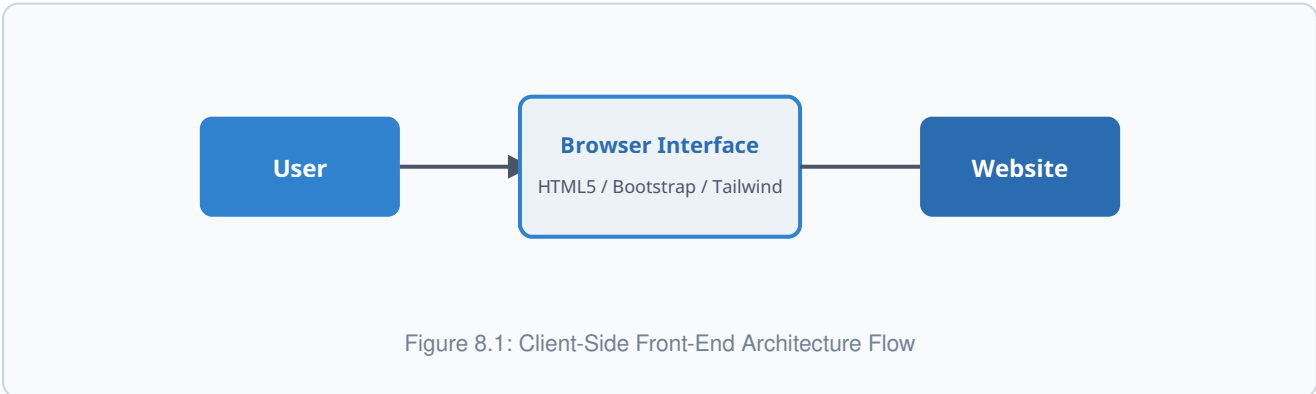
7. Non-Functional Requirements

Quality Attribute	Specification & Target standard
Security	Form inputs validated; simulated password hashing and safe session handling practices.

Quality Attribute	Specification & Target standard
Performance	Lightweight front-end design ensures minimal asset size and rapid DOM rendering speed.
Usability	Intuitive UI built with Bootstrap and Tailwind CSS, adhering to accessibility and visual contrast standards.
Reliability	Consistent client-side state handling and fallback interfaces for missing media or data.
Compatibility	Fully responsive across modern web browsers (Chrome, Firefox, Safari, Edge) and screen sizes.
Maintainability	Modular HTML/CSS structure with clean naming conventions for effortless future updates.

8. System Architecture

PlayMates follows a clean, client-centric front-end architecture. The presentation layer utilizes standard HTML5 structure combined with Bootstrap layout grids and Tailwind CSS utility styling for high responsive flexibility.



Architecture Note

This simplified mini-project is purposefully designed as a front-end Web Technology project; backend services (PHP/MySQL/XAMPP) are omitted from the active implementation stack to focus on front-end excellence.

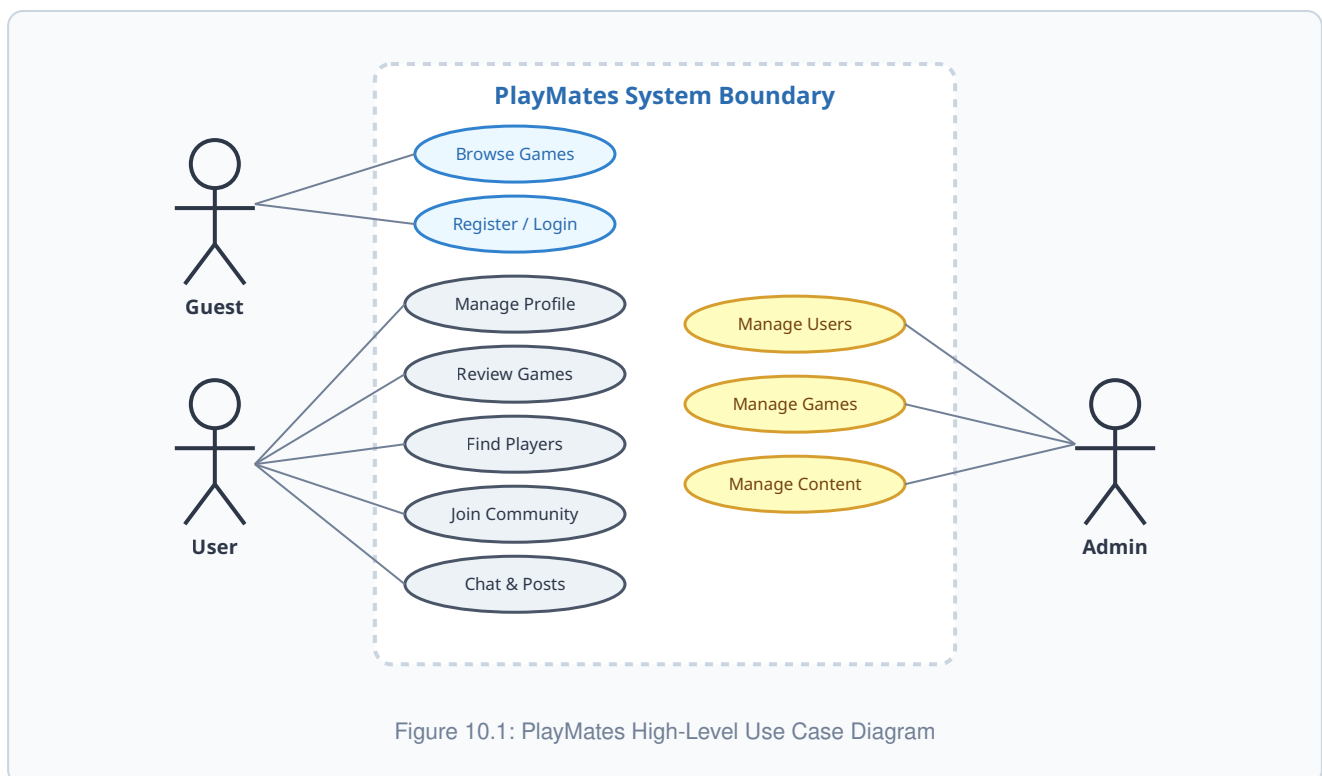
9. Database Model (Proposed Future Backend)

For the front-end project execution, client-side data structures are utilized. If a backend database is connected in future iterations, the following relational schema is proposed:

Table Name	Purpose & Stored Data
users	Stores user account details, credentials, region, and profile preferences.
games	Stores game details, titles, descriptions, genres, and cover images.
reviews	Stores user ratings (1-5 stars) and written text reviews per game.
communities	Stores gaming club details, rules, and creator user IDs.
community_members	Junction table mapping user memberships to specific communities.
posts	Stores user community feed posts, timestamps, and media references.
comments	Stores discussion comments linked to specific post IDs.
rooms	Stores active game lobby and room configuration data.
messages	Stores live chat messages sent within specific room IDs.
events	Stores esports and gaming event titles, schedules, and registration links.

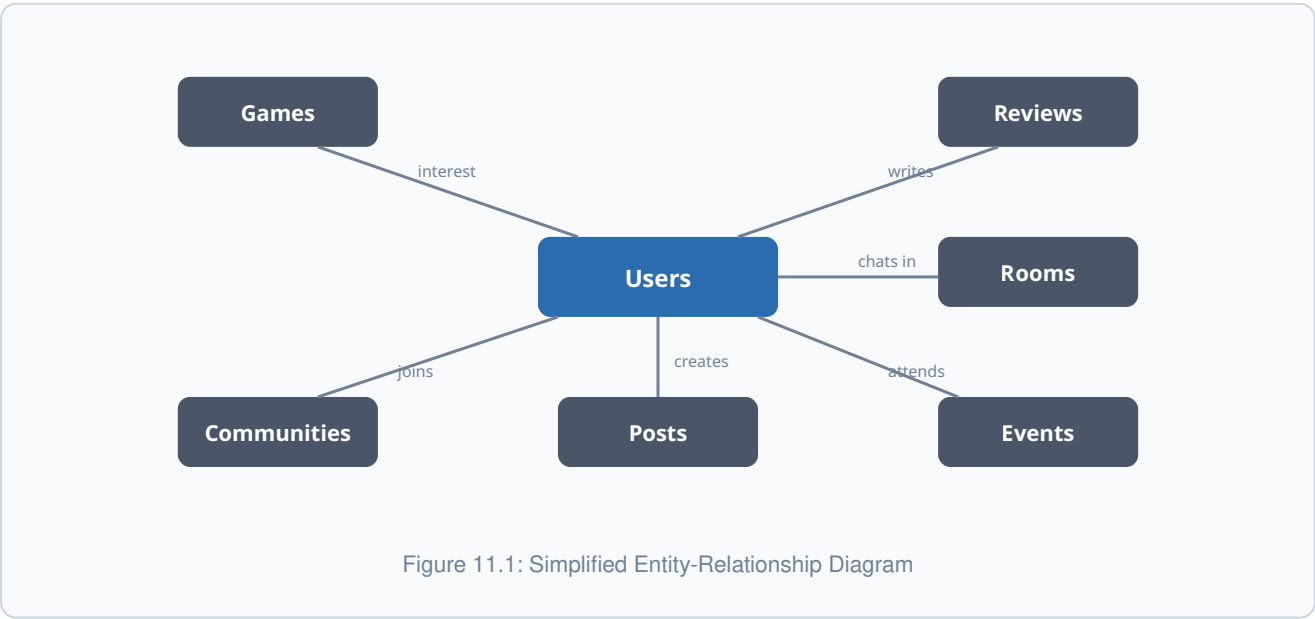
10. Use Case Diagram

The Use Case Diagram below illustrates the functional interactions available to Guests, Registered Users, and System Administrators within PlayMates.



11. Entity-Relationship (ER) Diagram

The ER diagram represents the core entities and logical relationships that structure the PlayMates data model.



12. Testing & Verification

System features have been systematically tested through front-end verification test cases to ensure correct UI responsiveness and functionality.

Test ID	Feature Under Test	Expected Output / Behavior	Status
TC-01	Registration Form	Accepts valid user inputs and validates required fields.	PASS
TC-02	Login Authentication	Valid credentials redirect user to authorized profile.	PASS
TC-03	Session Logout	Clears active session state and redirects to landing page.	PASS
TC-04	Game Search Filter	Dynamically updates displayed game cards based on input text.	PASS
TC-05	Review Submission	Appends new review entry dynamically to the game view.	PASS
TC-06	Star Rating System	Recalculates and renders average rating display correctly.	PASS
TC-07	Recommendations	Filter rules display matching genre games accurately.	PASS
TC-08	Community Join		PASS

Test ID	Feature Under Test	Expected Output / Behavior	Status
		Updates member count and updates user profile join state.	
TC-09	Game Room Chat	Appends chat message bubble into active conversation log.	PASS
TC-10	Event Join	Confirms registration state and updates event list entry.	PASS

13. Future Scope

The current implementation establishes a solid front-end core. Potential enhancements for subsequent project phases include:

- **Integrated Voice Chat:** WebRTC integration for real-time voice lobbies.
- **Mobile Application:** Responsive React Native or PWA extension for mobile devices.
- **Advanced Machine Learning Recommendations:** Collaborative filtering and personalized content recommendations.
- **External Game API Integration:** Live synchronization with Steam, Twitch, and RAWG APIs.
- **Friend & Direct Messaging System:** Private 1-on-1 messaging and friend list tracking.
- **Real-time Push Notifications:** WebSocket-powered alerts for comments, event start times, and room invites.

14. Conclusion

PlayMates provides an integrated platform designed to consolidate game discovery, player matchmaking, community reviews, and event coordination. The simplified architecture ensures a manageable scope for a Web Technology mini-project while delivering modern design, high usability, and clean code presentation suitable for college viva evaluations.

Project Summary Metadata

Final Technology Stack: HTML5, CSS3, Bootstrap, Tailwind CSS

University: MGM University

Project Lead Authors: Parth Chormale & Viraj Bhosale